# AgroLnk — Complete Production Demo & Testing Manual

**A Step-by-Step Interactive Guide for QA Testers & Stakeholder Demonstrations**

## 📋 Executive Overview

AgroLnk is an end-to-end B2B Agri-Trade Platform integrating **Spot Commerce, Live Reverse-English Auctions, 100% Escrow & 0.50% Commission, Multi-Axle Logistics, WDRA e-NWR Warehouses, 30-Day Working Capital Financing, Assayer Quality Inspections, and Anti-Circumvention Privacy Chat**.

This manual guides you step-by-step through every user role and business flow with exact account credentials, navigation paths, actions to perform, and expected results.

## 👥 1. Test Accounts & Role Access Matrix

| USER ROLE | PERSONA NAME | LOGIN EMAIL | PASSWORD | PRIMARY PORTAL ROUTE | KEY TESTING RESPONSIBILITIES |
|---|---|---|---|---|---|
| **Platform Admin** | Platform Admin | `admin@agrolnk.com` | `admin123` *(or any)* | `#/admin-dashboard` | 1-Click KYC Approvals, Escrow Ledger Auditing, Assayer Dispatch, Dispute Arbitration |
| **Verified Farmer** | Veerappan / Sakthi Vel | `farmer@agrolnk.com` | `farmer123` | `#/farmer-dashboard` | Spot Lot Creation, Live Auctions, e-NWR Warehouse Deposits, 75% LTV Loans |
| **Wholesale Buyer** | Maran S / Global Agro | `buyer@agrolnk.com` | `buyer123` | `#/buyer-dashboard` | Direct Spot Buying, 100% Escrow Funding, Live Bidding, Pre-Dispatch Assayer Requests |
| **Transporter** | Apex Fleet Logistics | `transporter@agrolnk.com` | `pass123` | `#/transporter-dashboard` | Multi-Axle Fleet Allocation, Corridor Checkpoint Updates, 6-Digit Delivery OTP |
| **Financier / NBFC** | Samunnati Capital | `financier@agrolnk.com` | `pass123` | `#/financier-dashboard` | 30-Day Credit Underwriting, Collateral Vault Liens, 1-Click Capital Disbursement |
| **Warehouse Operator** | Salem Agri Cold Hub | `warehouse@agrolnk.com` | `pass123` | `#/warehouse-dashboard` | 5,000 MT Storage Auditing, Multi-Chamber Temperature Zones, e-NWR Digital Titles |

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## 🧭 2. Step-by-Step Assistant Testing Scenarios

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### 🧪 SCENARIO 1: Identity KYC Gate & 1-Click Admin Approval

> **Goal:** Test that unverified accounts are prevented from posting trades until Admin verifies their identity documents.

- **Step 1.1 — Login as Farmer & Attempt Listing:**

        - **Role:** Farmer ( `farmer@agrolnk.com` )
        - **Navigation:** Open sidebar ➡️ Click **Post Produce** ( `#/farmer-create-listing` ).
        - **Action:** If unverified, observe the **Verification Required Modal** blocking unauthorized submission.
        - **Action:** Click **Submit KYC Documents** and upload an Aadhaar / GST document (Image or PDF).
    - **Step 1.2 — Switch to Platform Admin:**
        - **Role:** Admin ( `admin@agrolnk.com` )
        - **Navigation:** Go to **Admin Dashboard** ➡️ **KYC Verification Queue** ( `#/admin-verification` ).
        - **Action:** Locate the farmer in the pending queue.
        - **Action:** Click the 👁️ **View Document** button to preview the uploaded PDF/Image in the modal.
        - **Action:** Click the green ✓ **Approve** button.
    - **Step 1.3 — Verify Instant Privilege Grant:**
        - **Role:** Switch back to Farmer ( `farmer@agrolnk.com` ).
        - **Visual Check:** Notice the farmer now displays the emerald **Verified Producer** shield badge.
        - **Action:** Reopen **Post Produce**; the form is now 100% unlocked for trading.

✅ **Expected Outcome:** Secure role-gated onboarding where only KYC-approved producers can initiate commercial trades.

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## 🧪 SCENARIO 2: Spot Produce Listing & 0.50% Escrow Procurement

> **Goal:** Test listing creation and verify transparent 0.50% take-rate fee math (0.25% Buyer + 0.25% Farmer).

- **Step 2.1 — Farmer Creates a Produce Lot:**
    - **Role:** Farmer ( `farmer@agrolnk.com` )
    - **Navigation:** Go to **Post Produce** ( `#/farmer-create-listing` ).
    - **Form Values:**
        - **Commodity:** `Tomato`
        - **Variety:** `Hybrid Shivam`
        - **Grade:** `Grade A Premium`
        - **Quantity:** `500 kg` | **Unit Price:** `₹42 / kg`
        - **Location:** `Attur Farmgate Hub, Salem, Tamil Nadu`
    - **Action:** Click **Publish Listing**.
- **Step 2.2 — Buyer Browses & Places Escrow Order:**
    - **Role:** Wholesale Buyer ( `buyer@agrolnk.com` )
    - **Navigation:** Go to **Marketplace** ( `#/buyer-marketplace` ).
    - **Action:** Search for `Tomato` or filter by `Tamil Nadu` .
    - **Action:** Click on the listing card to view details ( `#/buyer-listing-detail` ).
    - **Action:** Click **Direct Purchase / Buy Now**.
- **Step 2.3 — Audit Fee Breakdown in Order Modal:**
    - **Quantity to Buy:** Enter `100 kg` .

- **Check Numbers:**
    - Commodity Subtotal: `100 kg × ₹42 = ₹4,200`
    - Buyer Platform Escrow Fee (0.25%): `₹10.50`
    - Total Buyer Deposit: `₹4,210.50`
- **Action:** Click **Confirm & Deposit to Escrow**.

✅ **Expected Outcome:** Order status moves to `confirmed` with funds securely held in escrow.

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## 🧪 SCENARIO 3: Pre-Dispatch Assayer Quality Check & Certification

> **Goal:** Test buyer quality protection where certified assayers inspect moisture and grade before dispatch.

- **Step 3.1 — Buyer Requests Assayer Inspection:**
    - **Role:** Wholesale Buyer (`buyer@agrolnk.com`)
    - **Navigation:** Go to **My Orders** (`#/buyer-orders`) and click on the newly placed tomato order.
    - **Action:** Locate the **Quality Inspection Desk** card inside the order summary modal.
    - **Action:** Click **Inspect Quality (Pre-Dispatch)**.
    - **Action:** Check parameters (`Moisture Content`, `Grade Assaying`) and click **Submit Quality Inspection Request**.
- **Step 3.2 — Admin Assayer Completes Physical Assay:**
    - **Role:** Platform Admin (`admin@agrolnk.com`)
    - **Navigation:** Go to **Dispute & Inspection Desk** (`#/admin-disputes`).
    - **Action:** Find the inspection request for Order #ORD-XXXX and click **Dispatch / Complete Assay**.
    - **Fill Assay Report:**
        - **Assay Grade:** `Grade A Premium`
        - **Moisture Content:** `10.8%` (Optimal)
        - **Foreign Matter:** `0.3%`
    - **Action:** Click **Approve & Dispatch Assay Report to Buyer**.
- **Step 3.3 — Buyer Views Certified Assay Sheet:**
    - **Role:** Switch back to Buyer (`buyer@agrolnk.com`).
    - **Visual Check:** Reopen the order. The inspection button is now replaced by a permanent **Readonly Certified Assay Report** showing all certified parameters.

✅ **Expected Outcome:** Zero quality risk for wholesale buyers through independent pre-dispatch certification.

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## 🧪 SCENARIO 4: Real-Time Live Reverse-English Auction Arena

> **Goal:** Test live bidding floor with real-time countdown, reserve price checks, and instant win settlement.

- **Step 4.1 — Farmer Launches Live Auction Floor:**

        - **Role:** Farmer ( `farmer@agrolnk.com` )
        - **Navigation:** Go to **Create Auction** ( `#/farmer-create-auction` ).
        - **Parameters:** `Basmati Rice (Pusa 1121)` | `20 Tonnes` | Starting Bid: `₹38 / kg` | Reserve: `₹45 / kg` .
        - **Action:** Click **Start Live Auction Floor**.
- **Step 4.2 — Buyer Enters Arena & Places Competitive Bid:**
    - **Role:** Wholesale Buyer ( `buyer@agrolnk.com` )
    - **Navigation:** Go to **Live Auctions** ( `#/buyer-live-auctions` ) ➡️ Click **Enter Auction Arena** ( `#/auction-room` ).
    - **Action:** Place a bid above reserve price (e.g. `₹46 / kg` ).
    - **Visual Check:**
        - Live bid log records the bid timestamp and bidder alias.
        - Status transitions from `Reserve Not Met` to `Reserve Met ✓` in emerald green.
        - Real-time timer decrements towards auction close.

✅ **Expected Outcome:** Dynamic, competitive price discovery with automated win conversion into an escrow contract.

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## 🍼 SCENARIO 5: Multi-Axle Logistics Manifest & Delivery Confirmation OTP

> **Goal:** Test corridor GPS milestone tracking and secure milestone-based escrow fund release.

- **Step 5.1 — Transporter Updates Corridor Milestones:**
    - **Role:** Transporter ( `transporter@agrolnk.com` )
    - **Navigation:** Go to **Logistics Corridor Hub** ( `#/transporter-dashboard` ).
    - **Action:** Select the assigned freight lot.
    - **Action:** Assign Truck Type: `Multi-Axle 16T Reefer (TN-28-AC-4091)` .
    - **Action:** Advance transit status: `Picked Up` ➡️ `In Transit` ➡️ `Arrived at Hub` .
- **Step 5.2 — 6-Digit Delivery Confirmation OTP:**
    - **Role:** At the delivery location, Buyer shares the **6-digit delivery OTP**.
    - **Action:** Transporter enters OTP and clicks **Verify & Complete Delivery**.
- **Step 5.3 — Verify Automatic Escrow Release:**
    - **Role:** Platform Admin ( `admin@agrolnk.com` ) ➡️ **Escrow Ledger** ( `#/admin-escrow` ).
    - **Visual Check:** Order updates to `Completed / Escrow Released to Farmer` with the exact 0.50% fee retained by AgroLnk.

✅ **Expected Outcome:** 100% fraud-proof settlement; farmer receives payout only after verified delivery.

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## 🧪 SCENARIO 6: Anti-Circumvention Privacy Chat & Smart Evasion Shield

> **Goal:** Test that contact evasion tactics (split phone digits, domain keywords, number words) are blocked in real time.

- **Step 6.1 — Open Direct Privacy Channel:**
  - **Role:** Wholesale Buyer ( `buyer@agrolnk.com` ).
  - **Navigation:** Open **My Orders** ( `#/buyer-orders` ) ➡️ Click the 💬 **Privacy Chat** drawer button.
  - **Action:** Select channel: **Veerappan (Farmer) • Direct Channel**.
- **Step 6.2 — Test Circumvention Attack Vectors:**
  - Try typing the following messages to test the smart shield:

| TEST INPUT SCENARIO | MESSAGE CONTENT TYPED | EXPECTED SMART SHIELD MASKING |
|---|---|---|
| Standard Phone Number | `Call me at 9840123456 for direct deal` | `Call me at 98*** ***56 [Protected Phone] for direct deal` |
| Email Address | `Send invoice to buyer.agro@gmail.com` | `Send invoice to [Protected Email]` |
| Domain Fragment | `Contact me on yahoo or dot com` | `Contact me on [Protected Domain] or [Protected Domain]` |
| Split Number Evasion (Sliding Window) | Msg 1: `98755`<br>Msg 2: `67890` | **Sliding Window Intercepts:** Masked to `98*** [Protected Fragment]` + **Security Warning Bot Alert** triggered! |
| Number Words | `call me at nine eight four zero...` | Words normalized to digits and masked safely. |

- **Step 6.3 — Verify Real-Time Bidirectional Sync:**
  - **Role:** Login as Farmer in another tab/browser ( `farmer@agrolnk.com` ).
  - **Visual Check:** Both parties see the synchronized conversation thread in real time.

✅ **Expected Outcome:** Zero PII leakage; traders communicate freely while platform revenue is safeguarded.

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## 🧪 SCENARIO 7: e-NWR Warehouse Deposits & 30-Day Working Capital Financing

> **Goal:** Test electronic Negotiable Warehouse Receipts (e-NWR) and 75% LTV commodity-backed credit underwriting.

- **Step 7.1 — Farmer Deposits Produce into Cold Storage:**
  - **Role:** Farmer ( `farmer@agrolnk.com` ) ➡️ **e-NWR Vault** ( `#/farmer-inventory` ).
  - **Action:** Click ➕ **Deposit Produce**, select `Salem Agri Cold Hub` , `1000 kg Tomato` , `Chamber B4` .
  - **Visual Check:** An electronic title `eNWR-2024-XXXX` is minted.
- **Step 7.2 — Farmer Applies for 75% LTV Working Capital:**

- **Action:** Click **Apply for Working Capital** on the receipt card.
- **Action:** Request `₹25,000` (Max ceiling: `75% of ₹42,000 = ₹31,500`) for `30 Days`.
- **Action:** Click **Submit Working Capital Request**.

- **Step 7.3 — NBFC Financier Approves & Places Collateral Lien:**
  - **Role:** Financier (`financier@agrolnk.com`) ➡️ **Underwriting Desk** (`#/financier-underwriting`).
  - **Action:** Click **Review & Underwrite** ➡️ Click **Approve & Disburse Funds**.
  - **Visual Check:** Open **Collateral Vault** (`#/financier-collateral-vault`). The receipt is tagged with an active **Lien / Pledge Badge**, locking it from unauthorized withdrawal until repaid.

✅ **Expected Outcome:** Immediate working capital liquidity for farmers without predatory middleman loans.

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## 💉 SCENARIO 8: Live Escrow Account Gateway API & Realtime Webhooks

> **Goal:** Test the virtual nodal Escrow Account engine, simulated bank gateway webhooks, and live balance movements in real time.

- **Step 8.1 — Open the Live Escrow Gateway Console:**
  - **Role:** Platform Admin (`admin@agrolnk.com`)
  - **Navigation:** Go to **Admin Dashboard** (`#/admin-dashboard`) or **Escrow Ledger** (`#/admin-escrow`).
  - **Action:** Click the green ⚡ **Live Escrow API** button in the header.
  - **Visual Check:** The **Live Escrow Account Gateway & Nodal API** modal opens displaying:
    - **Nodal Partner:** `ICICI Bank Ltd. (Institutional Agri-Nodal Branch)`
    - **Trustee Partner:** `Vistra / Axis Trustee Services Ltd. (SEBI Reg.)`
    - **Virtual Nodal Account:** `992010884210982` • IFSC: `ICIC0000104`
    - **Live Metrics:** Locked in Escrow, Settled Payouts, 0.50% AgroLnk Revenue pool.

- **Step 8.2 — Test Inward Escrow Deposit Simulator:**
  - **Navigation:** Switch to the ⚡ **Live Interactive Simulator** tab in the modal.
  - **Action:** Enter Commodity: `Tomato`, Trade Amount: `₹5,000`, Buyer: `Maran S`, Farmer: `Veerappan`.
  - **Action:** Click ⚡ **Trigger Live Inward Deposit**.
  - **Visual Check:**
    - Instant toast confirmation: `✓ Captured ₹5,012.50 into Escrow Vault! UTR: CMSICICI...`.
    - Locked vault balance increments immediately.

- **Step 8.3 — Test OTP Milestone Payout & Realtime Webhook Bus:**
  - **Action:** Click 🔐 **Verify OTP & Disburse to Farmer**.
  - **Visual Check:** Funds shift from **Locked in Escrow** to **Settled Payouts**, and the 0.50% commission pool increments in real time!
  - **Action:** Switch to the 📡 **Live Webhook Logs** tab to view the live JSON event bus:
    - `webhook.escrow.payment_captured`
    - `webhook.escrow.payout_disbursed`

✅ **Expected Outcome:** Full institutional escrow transparency with real-time auditability and zero settlement leakage.

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## 🛡️ 3. Quick Sanity & Production Readiness Checklist

- ☑ **Zero Build Errors:** `npm run build` compiles 1,948+ modules cleanly with 0 errors.
- ☑ **Zero SQL Policy Collisions:** Supabase schema uses idempotent drop/create definitions.
- ☑ **Live Nodal Escrow API:** Real-time ledger, simulated webhooks, and virtual account routing operational.
- ☑ **Responsive UI:** Tested across Desktop, Tablet, and Mobile viewports.
- ☑ **Channel Deduplication:** Chat displays 1 entry per business partner regardless of order volume.
- ☑ **Accurate Revenue Math:** 0.25% Buyer + 0.25% Farmer = 0.50% Total Platform Take-Rate.

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